

MD. NOMAN BISWAS SIBLY

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🌐 Portfolio Website —  LinkedIn —  GitHub —  Scholar

📍 Shahid President Ziaur Rahman Hall, RUET, Kazla, Rajshahi-6204, Bangladesh

EDUCATION

Rajshahi University of Engineering & Technology (RUET)

Dec 2022 – Present

B.Sc. in Electrical & Computer Engineering

CGPA: 3.56/4.00

(after 6th semester)

BAF Shaheen College, Jashore

Jul 2019 – Dec 2021

HSC – Science

GPA: 5.00/5.00

RESEARCH

Publications

- First Author, “Which Reliability Signal to Trust? Output vs. Representation Space Uncertainty Under Distribution Shift in Pathology Foundation Models”, accepted at *MI4MedFM Workshop, MICCAI 2026*. [\[Code\]](#)
- Second Author, “A Comprehensive Review of Convolutional Neural Networks: Foundations, Enhancements, and Applications”, *Neural Computing and Applications* (Springer, Q1, IF 5.102), 38(4):56, 2026. [\[DOI\]](#)
- First Author, “Noise-Robust Gallbladder Ultrasound Image Classification Using U-Net GAN Denoising and Vision Transformer”, *IEEE QPAIN 2026*. [\[DOI\]](#)

Manuscript in Preparation

- First Author, “JTADC-Net: An Adversarial Joint Task-Aware Denoising and Classification Network for Noise Robust Gallbladder Disease Diagnosis” — joint adversarial denoising and classification for noise-robust gallbladder diagnosis. [\[Code\]](#)

Ongoing Research

- Trustworthy AI in Healthcare (Thesis)

EXPERIENCE

Research Assistant (Part-time, Remote)

2026 – Present

ELITE Research Lab

Bangladesh

- Working on mechanistic interpretability of neural networks to make predictions explainable.

AI & ML Trainee

2025 – 2026

BDSET

Bangladesh

- Applied ML training: preprocessing, model building, tuning, evaluation (Python, TensorFlow, Scikit-learn).

Programming Mentor

2024 – 2025

RUET Analytical Programming Lab

Rajshahi

- Taught algorithms/debugging and guided students in competitive programming and contest preparation.

PROJECTS

- ↔ **JTADC-Net: Noise-Robust Gallbladder Disease Classification:** Developed a GAN-assisted ViT framework for robust gallbladder ultrasound classification under noisy conditions, packaged as a real-time medical image classification web app with deployment support. [\[Code\]](#) [\[Demo\]](#) [2025]
- ↔ **TruthLayer-Gemma4: Trustworthy Vision-Language AI:** Explored reliability, hallucination reduction, and verification strategies in multimodal vision-language systems, investigating trustworthy AI methods for safer and more interpretable inference. [\[Code\]](#) [2025]
- ↔ **ATmega Neural Network – Digit Recognizer:** Built a lightweight neural network for digit classification on ATmega microcontrollers using optimized integer computations, with memory-efficient TinyML inference pipelines for constrained embedded environments. [\[Code\]](#) [2025]

PROBLEM SOLVING

[Codeforces](#) (Pupil, max 1263; 1186 solved) · [LeetCode](#) (125 solved) · [VJudge](#) (378 solved)

CERTIFICATES & ACHIEVEMENTS

CS50: Introduction to Computer Science <i>Harvard University</i>	2023 <i>Verify</i>
· Learned Python, algorithms, data structures, and problem-solving	
CS50: Databases with SQL <i>Harvard University</i>	2024 <i>Verify</i>
· Covered SQL, schemas, indexing, transactions, triggers, and database design	
Machine Learning Specialization (Andrew Ng) <i>DeepLearning.AI & Stanford</i>	2024 <i>Verify</i>
· Completed 3-course ML specialization (Supervised ML, Advanced ML, ML Engineering)	
Machine Learning with Python <i>IBM & Coursera</i>	2024 <i>Verify</i>
· Completed ML workflow: preprocessing, training, tuning, and evaluation	
2nd Runner-Up – ECE Youth Programming Contest <i>RUET</i>	2023
· Ranked 3rd in the department-level competitive programming contest	

TECHNICAL SKILLS

- ↔ **Languages:** C, C++, Python, JavaScript, SQL
- ↔ **Data Science & ML Tools:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib
- ↔ **Web Development:** HTML, CSS, Django, MySQL
- ↔ **Other Tools:** Git, GitHub, Linux, Jupyter Notebook, Data Analysis